

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$, yet trained slopes fall short (Figure 4). Even though larger widths admit smaller λ at comparable construction error (Figure 2b), small slopes can delay readout learning by weakening target-relevant kernel directions. We quantify this delay by replacing Section 3.1’s direct least-squares solve with GD on fixed features. This connects prior analysis of spectral bias (Xu et al., 2020; Tancik et al., 2020) to readout optimization in the quasi-interpolant dictionary.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$, let $\mu_1 \geq \mu_2 \geq \dots \geq 0$ be the normalized readout-kernel eigenvalues. For a nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, p_i is the fraction of sampled target energy projected onto kernel eigenvector i , and $\mu_i/\mu_1 \leq \bar{\rho}_i(\gamma) \leq 1$. The gamma-dependent upper bounds $\bar{\rho}_i$ come from the spectral Theorem A.1 in Appendix A.

Proof sketch. The readout kernel contains normalized inner products of feature vectors at pairs of training points. To bound its spectrum, we separate the constant sample direction and approximate the sum over centers by an integral. This approximation depends explicitly on $d \coth(\gamma d)$, where d separates those points. Bounding the finite-center corrections gives $\bar{\rho}_i(\gamma)$; standard GD eigenmode decay then yields the output-error bound.

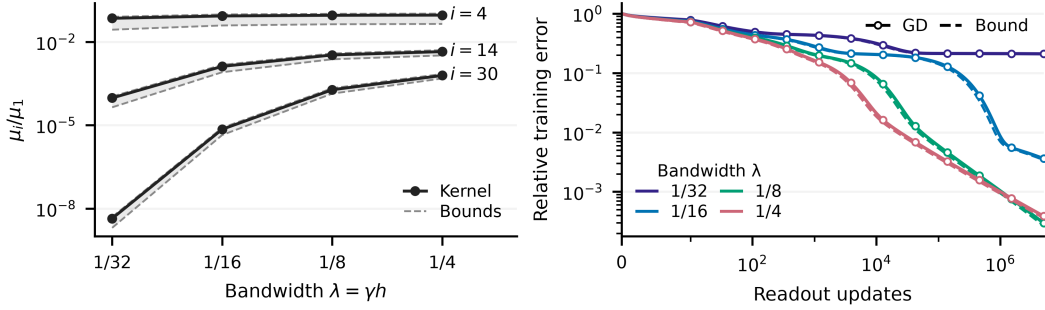


Figure 3: **Small slopes weaken kernel directions and delay readout learning.** Width $W = 512$; target $f(x) = [\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$ on $[-1, 1]$. **Left:** Kernel ratios μ_i/μ_1 (dots) and intervals from Theorem A.1 (dashed boundaries, shading) for three ranks carrying substantial target energy. Lines connect ordered ranks; eigenvectors change with bandwidth. **Right:** Executed frozen-readout GD errors (solid, hollow markers) and square roots of Theorem 3.2’s lower bounds (dashed). The error bound combines theorem-derived rate bounds with all resolved target projections onto the frozen kernel’s eigenvectors. Errors are relative training RMS through five million updates; geometry and protocol are in Appendix B.

The spectral bounds of Theorem A.1 track how bandwidth changes target-relevant ratios μ_i/μ_1 (Figure 3, left). Since μ_1 limits a stable GD step, the tiny ratios at small bandwidths leave needed target components slow to decay. Theorem 3.2 converts these rate bounds into output-error lower bounds that closely track executed GD (Figure 3, right): small slopes delay accuracy already available in the dictionary.

Joint training shows an empirical acquisition shortfall under both GD and Adam: the population’s mean bandwidth remains below the supplied construction scale, with a substantial output-error gap over the same training budget (Figure 4). We next examine why training fails to acquire useful slopes and centers together.

3.5 TRAINING MUST ACQUIRE USEFUL SLOPES AND CENTERS

Joint training also leaves a substantial precision gap over this budget. After five million updates, joint Adam reaches median dense-grid relative error 1.81×10^{-3} , versus 6.62×10^{-7} on the supplied $\lambda = 1/4$ features (Figure 4, left).

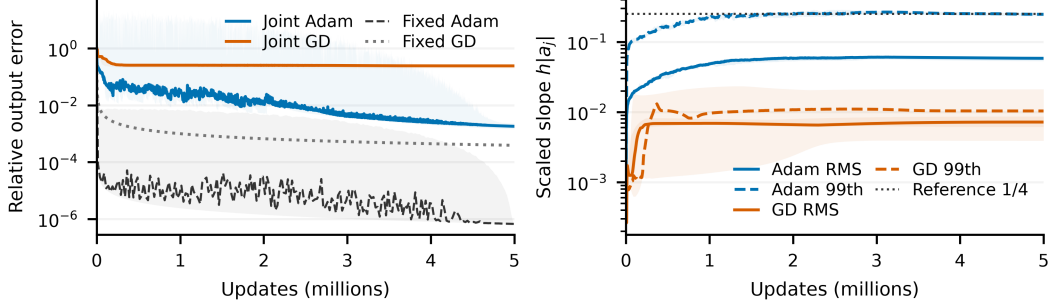


Figure 4: **Partial slope acquisition leaves a precision gap.** **Left:** Joint Adam/GD relative output errors and frozen uniform $\lambda = 1/4$ references under the same five-million-update budget. **Right:** RMS and 99th-percentile scaled slopes $h|a_j|$ in the same joint runs. The reference is a supplied geometry, not a necessary slope threshold. Joint curves summarize five paired seeds; shading retains seed variation and recorded within-bin extrema. Cosine schedules span the full horizon. These are empirical comparisons; Appendix B gives the protocol.

The population’s mean bandwidth remains below the supplied reference (Figure 4, right). Refitting the readout on the learned features barely improves their error (Appendix B), so unfinished readout optimization alone does not explain the gap. The next analysis therefore concerns the joint acquisition of useful slopes and centers. The construction supplies both this geometry and the readout directly.